

VEHICLE RENTAL MANAGEMENT SYSTEM

Final Project Report

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Course: Database Management Systems

Project Title: Vehicle Rental Management System

Project Category: Transportation (Vehicle Rental)

Database: Oracle Database 21c Express Edition

Tools Used:

Oracle SQL Developer

Oracle Database 21c Express Edition

Draw.io (Diagrams.net)

GitHuB

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1.Introduction

The Vehicle Rental Management System is a database application developed to automate the daily operations of a vehicle rental company. The system helps employees manage customers, vehicles, rentals, payments, and vehicle returns efficiently.

This project was developed using Oracle Database 21c Express Edition and Oracle SQL Developer. It demonstrates the use of database design, SQL, PL/SQL programming, normalization, triggers, procedures, functions, packages, and business process modeling.

The main goal of the system is to improve data accuracy, reduce manual work, and provide a secure and reliable database solution for vehicle rental management.

2.Projection Description

The Vehicle Rental Management System is designed to automate and simplify the management of vehicle rental operations. It allows employees to register customers, manage vehicle information, process rental transactions, record payments, and track vehicle returns.

The system stores all data in an Oracle database, ensuring accuracy, consistency, and security. It reduces paperwork, minimizes human errors, and improves the efficiency of daily operations.

The database also includes audit logging, security triggers, and additional features such as views, indexes, and sequences to improve performance and maintain data integrity.

3.Problem Statement

Many vehicle rental companies still rely on manual methods or spreadsheets to manage customer information, vehicle records, rental transactions, and payments. These traditional methods often lead to data duplication, errors, loss of information, and inefficient management.

To solve these problems, a computerized database system is required. The Vehicle Rental Management System provides a secure and efficient solution by automating rental operations, maintaining accurate records, improving data security, and reducing manual work.

4.objectives

General Objective

The main objective of this project is to develop a Vehicle Rental Management System using Oracle Database to automate rental operations, improve data management, and ensure data security.

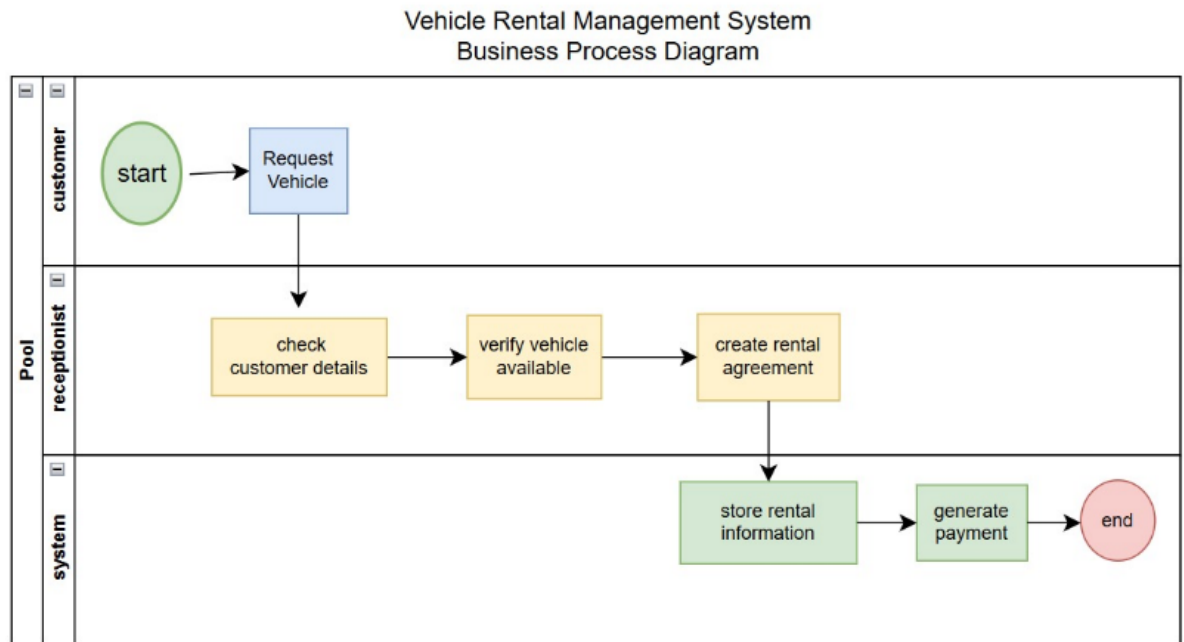
Specific Objectives

- To register and manage customer information.
- To store and manage vehicle records.
- To process vehicle rental transactions.
- To record customer payments.
- To manage vehicle returns.
- To implement SQL and PL/SQL features such as procedures, functions, packages, and triggers.
- To improve database performance using views, indexes, and sequences.
- To ensure data integrity through normalization up to Third Normal Form (3NF).

5.Business Process Diagram

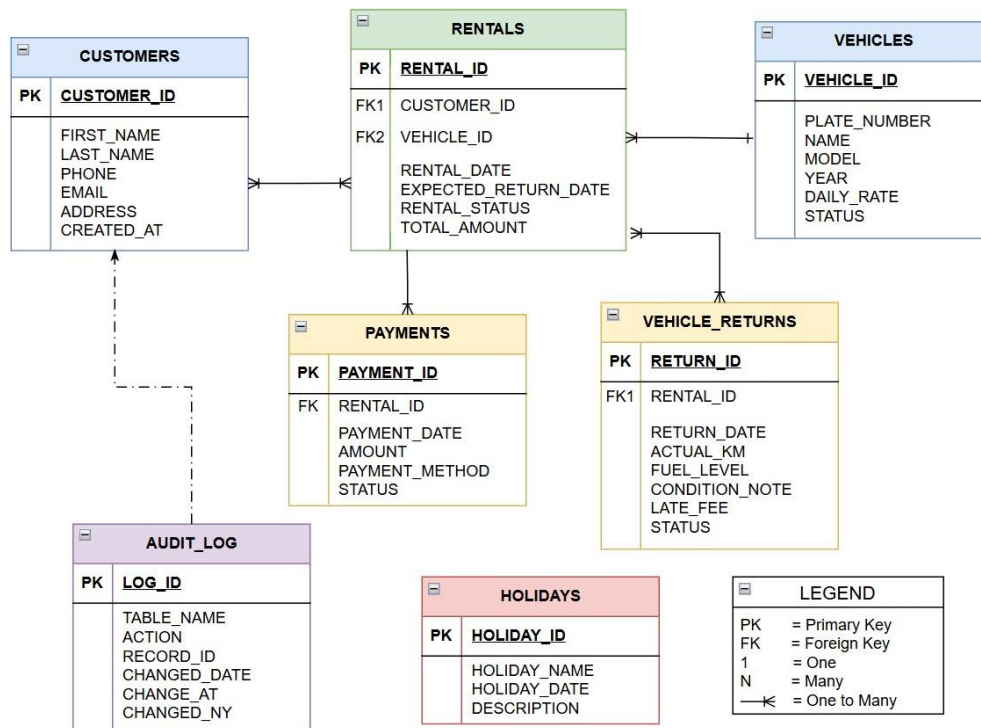
The following Business Process Diagram illustrates the workflow of the Vehicle Rental Management System. It shows the interaction between the customer, receptionist,

payment process, and vehicle return process.



6. Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the structure of the database. It shows the entities, their attributes, and the relationships between customers, vehicles, rentals, payments, vehicle returns, holidays, and the audit log.



7.Database Normalization

First Normal Form (1NF)

The database satisfies the First Normal Form (1NF) because each table contains atomic values, each record is unique, and every table has a primary key.

Second Normal Form (2NF)

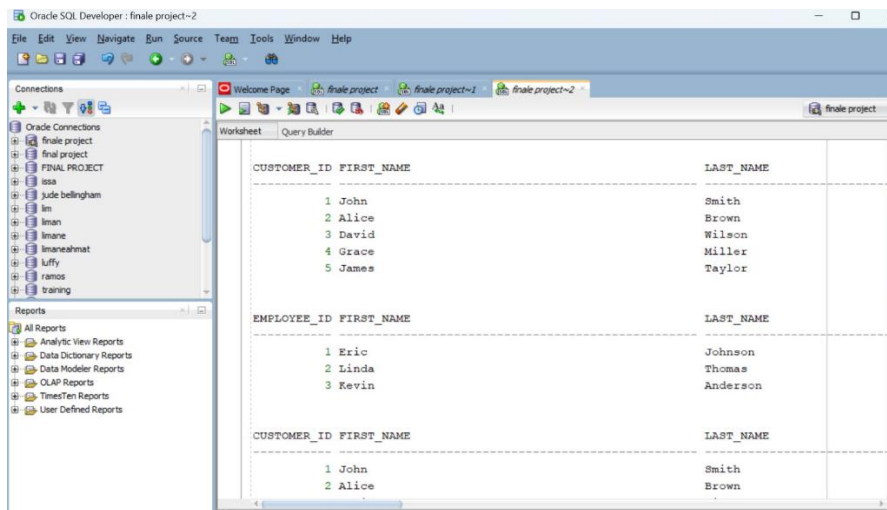
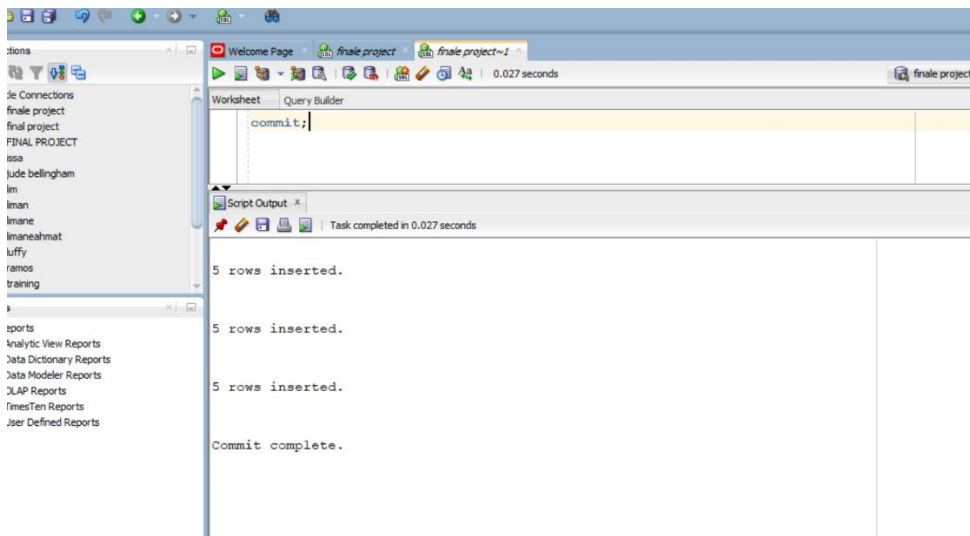
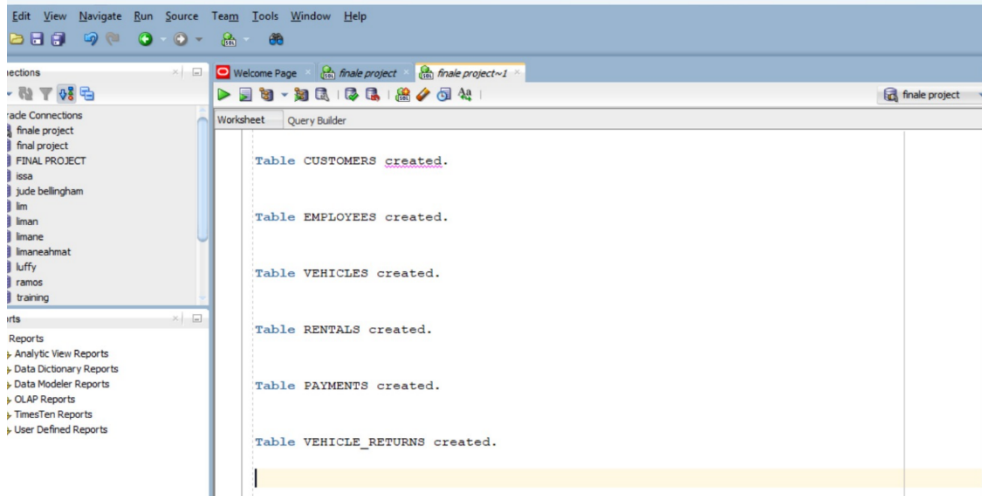
The database satisfies the Second Normal Form (2NF) because all non-key attributes depend entirely on the primary key. There are no partial dependencies.

Third Normal Form (3NF)

The database satisfies the Third Normal Form (3NF) because there are no transitive dependencies. Customer, vehicle, rental, payment, vehicle return, holiday, and audit information are stored in separate related tables, reducing redundancy and improving data integrity.

8.Database Implementation

The database was implemented using Oracle Database 21c Express Edition and Oracle SQL Developer. The implementation included creating tables, defining primary and foreign keys, applying constraints, and inserting sample data. All SQL statements were executed successfully.



1 Eric	Johnson
2 Linda	Thomas
3 Kevin	Anderson

VEHICLE_ID	BRAND	MODEL
1	Toyota	Corolla
2	Honda	Civic
3	Nissan	X-Trail
4	Hyundai	Tucson
5	Ford	Ranger

RENTAL_ID	CUSTOMER_ID	EMPLOYEE_ID	VEHICLE_ID	RENTAL_DA	RETURN_DA	TOTAL_AMOUNT
1	1	1	1	01-JUL-26	05-JUL-26	250
2	2	2	2	02-JUL-26	06-JUL-26	240
3	3	1	3	03-JUL-26	07-JUL-26	320
4	4	3	4	04-JUL-26	08-JUL-26	360

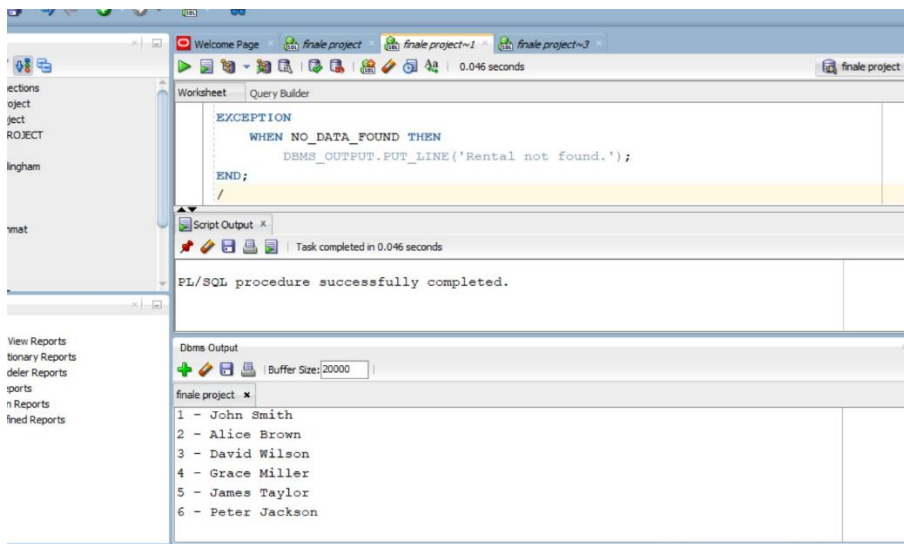
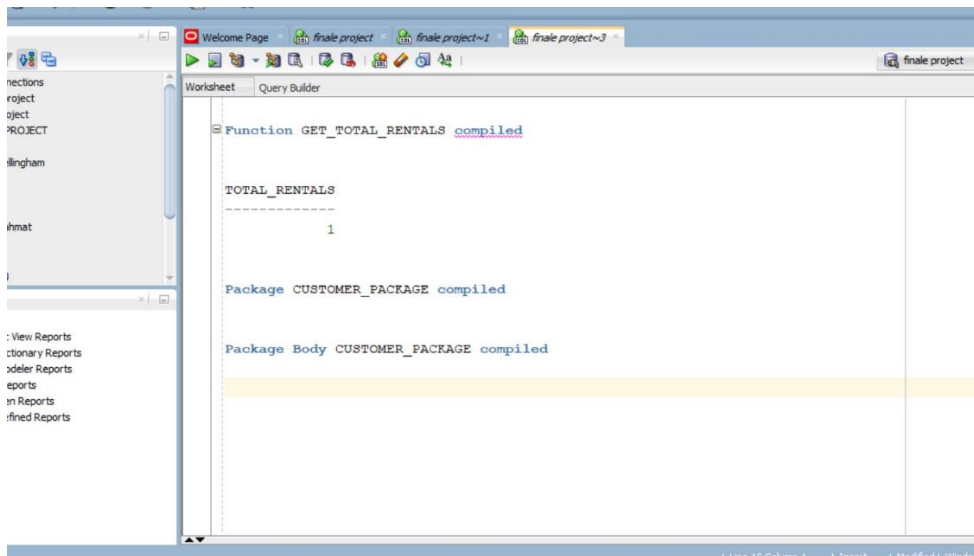
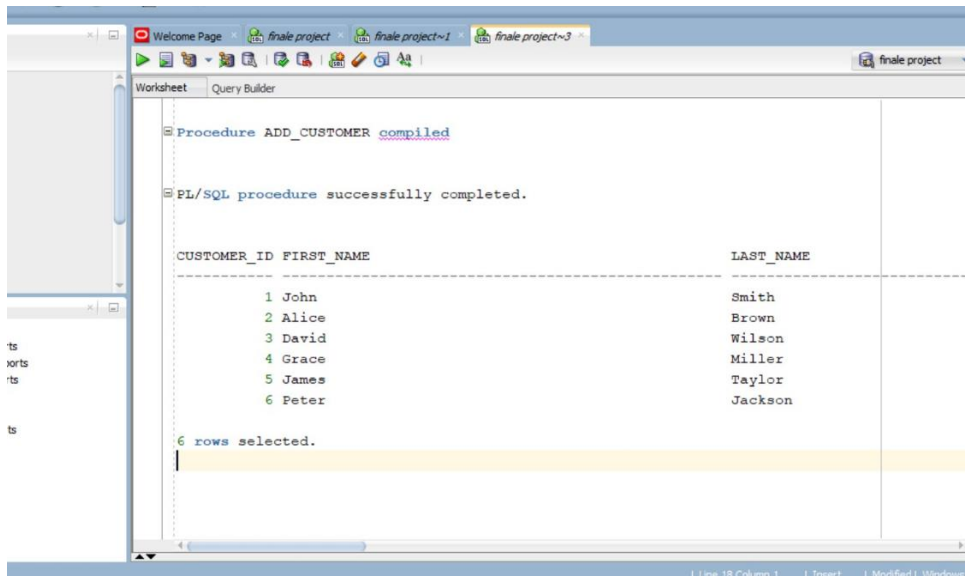
4 Grace	Miller
5 James	Taylor

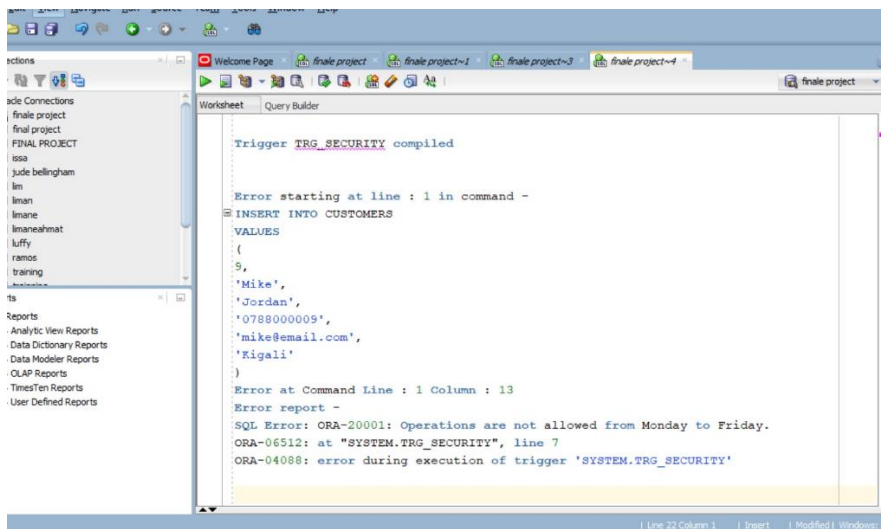
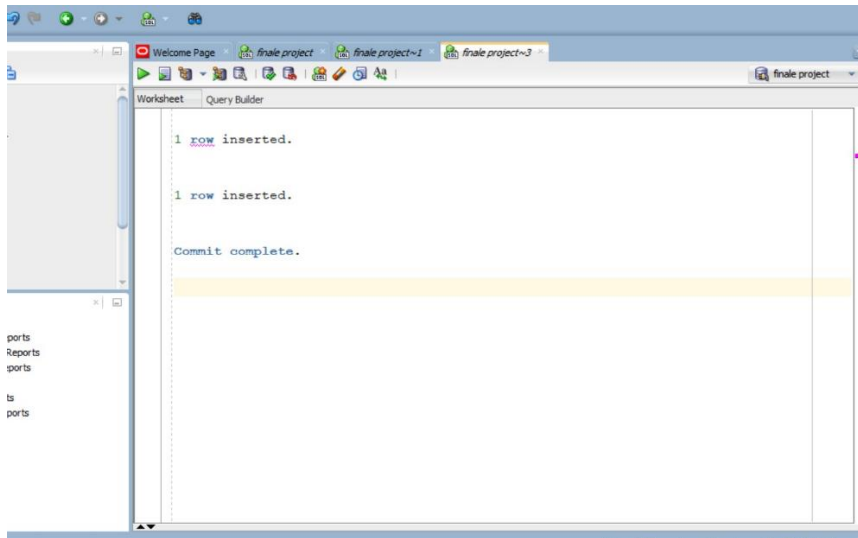
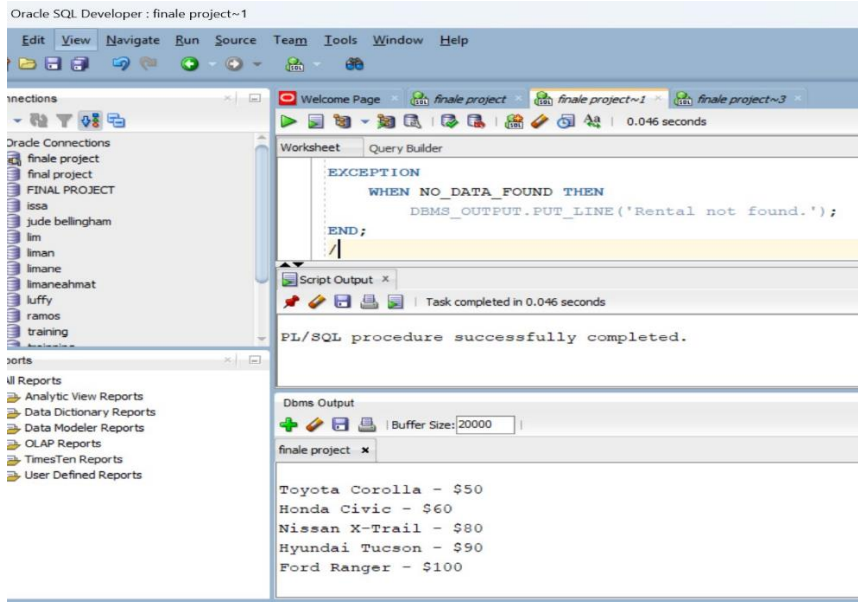
PAYMENT_ID	AMOUNT	PAYMENT_METHOD	RENTAL_ID
1	250	Cash	1
2	240	Credit Card	2
3	320	Mobile Money	3
4	360	Bank Transfer	4
5	400	Cash	5

RETURN_ID	ACTUAL_RE	CONDITION_STATUS	REMARKS
1	05-JUL-26	Excellent	Returned on time
2	06-JUL-26	Good	No damage
3	07-JUL-26	Good	Clean vehicle
4	08-JUL-26	Fair	Minor scratches
5	09-JUL-26	Excellent	Perfect condition

9. SQL and PL/SQL Features

The project uses several SQL and PL/SQL features to improve the functionality of the database. These features include stored procedures, functions, packages, triggers, cursors, and exception handling. They automate operations, improve data security, and simplify database management.



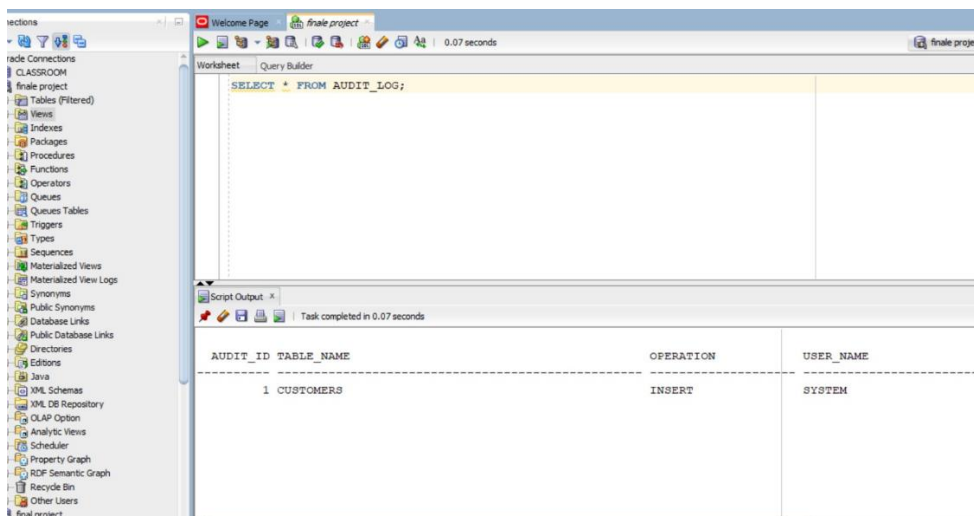
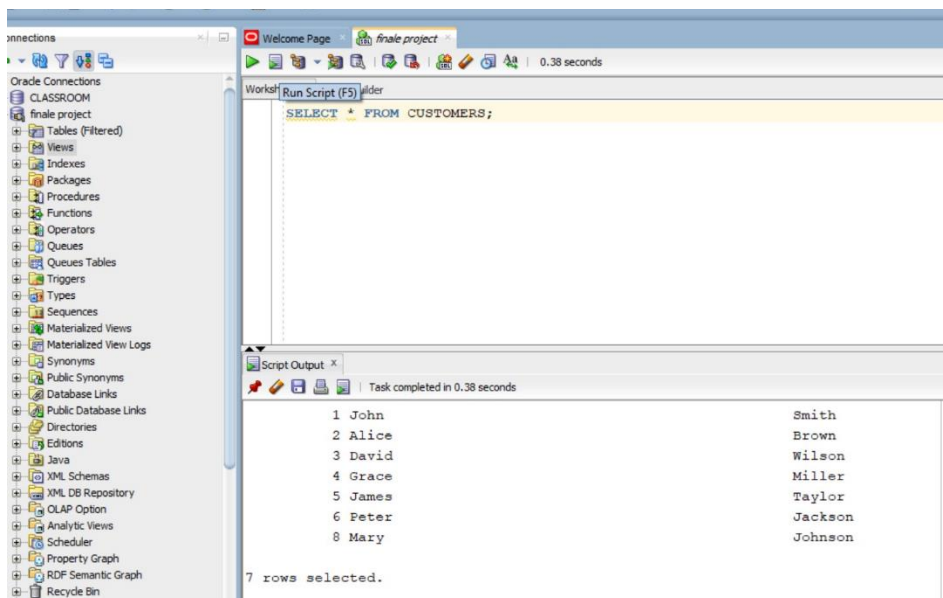


10.Innovation

To improve the performance and functionality of the database, several advanced Oracle features were implemented. A VIEW was created to simplify data retrieval, an INDEX was added to improve query performance, and a SEQUENCE was used to generate unique identifiers automatically. These features make the database more efficient, reliable, and easier to maintain.

11.Testing

The database was tested by executing SQL and PL/SQL statements. Insert, update, delete, and select operations were performed successfully. The stored procedure, function, package, trigger, cursor, and exception handling were also tested and produced the expected results.



12.Conclusion

The Vehicle Rental Management System was successfully designed and implemented using Oracle Database 21c Express Edition and Oracle SQL Developer. The project demonstrates the use of SQL and PL/SQL programming to manage customer information, vehicles, rentals, payments, and returns. The system improves efficiency, reduces manual work, and ensures data integrity through a well-designed database structure. This project also enhanced practical knowledge of Oracle database development.

13.References

1. Oracle Database 21c Express Edition Documentation.
2. Oracle SQL Developer Documentation.
3. Oracle PL/SQL Language Reference.
4. Database Management Systems Course Notes.
5. Draw.io (diagrams.net) Documentation.